

1 Other First Order Methods



Mathematics
and Statistics

$$\int_M d\omega = \int_{\partial M} \omega$$

Mathematics 2C03

Introduction to Differential Equations

Instructor: David Earn

Fall 2026

Announcements

- The TA is Illya Kierkosz (kierkosi@mcmaster.ca)
 - Office hours by appointment: e-mail to arrange a meeting.
- Course web site: <https://davidearn.github.io/math2c/>
- The course outline is on the [Simple Syllabus](#) web site.
- [Avenue to Learn](#) provides secure access to Echo360 live streams and lecture recordings.
- Assignments and class polls are on [childsmath](#).
- **e-mail:** PLEASE always include 2C03 in the subject line of any message about this course, and PLEASE indicate what the issue is. For example:
 - Subject: Math 2C03: confusion about assignment 1
 - Subject: Math 2C03: conflict with test window
- Office hour: Thursdays, 9:30–10:20 am, after class.

Test 1: action required

TEST 1

ACTION REQUIRED

[Course tests page](#)

- 1 **Read** the Flexible Testing Student Sign-Up and Testing Guide on that page.
- 2 **Choose** one of the Test 1 writing times in the table; sign up when booking opens.

Bernoulli equations

Bernoulli equations

A *Bernoulli equation* has the form

$$\frac{dy}{dx} + p(x)y = f(x)y^r, \quad r \neq 0, 1.$$

The excluded cases $r = 0$ and $r = 1$ reduce to *first-order linear ODEs*, which we have already solved using EMSID. For $r \neq 0, 1$, it is nonlinear because of y^r .

Replacing the RHS by zero gives the *complementary linear equation*:

$$\frac{dy}{dx} + p(x)y = 0,$$

As before, find a nonzero solution y_1 of this equation, then try

$$y = u(x)y_1(x),$$

i.e., let's try *the same variation of parameters technique*.

Bernoulli equations

Substitute $y = uy_1$ into $\frac{dy}{dx} + p(x)y = f(x)y^r$:

$$\begin{aligned}y_1 \frac{du}{dx} + u \frac{dy_1}{dx} + p(x)uy_1 &= f(x)(uy_1)^r \\y_1 \frac{du}{dx} &= f(x)y_1^r u^r \quad (*)\end{aligned}$$

since $\frac{dy_1}{dx} + p(x)y_1 = 0$.

For $u \neq 0$, this is *separable*:

$$u^{-r} \frac{du}{dx} = f(x)y_1^{r-1}.$$

Before dividing by u^r : check whether $u \equiv 0$, hence $y \equiv 0$, is also a solution.

Bernoulli equations

Example (Solve $\frac{dy}{dx} - y = xy^2$)

The complementary linear equation is $\frac{dy}{dx} - y = 0$. By EMSID, $P(x) = -1$, so the integrating factor is e^{-x} . Thus $\frac{d}{dx}(e^{-x}y) = 0$ gives $y = ce^x$. Take $y_1 = e^x$.

Set $y = ue^x$. The ODE (*) for u on the preceding slide becomes

$$\begin{aligned}e^x \frac{du}{dx} &= xu^2 e^{2x} \\ \frac{1}{u^2} \frac{du}{dx} &= xe^x, \quad (u \neq 0). \\ -\frac{1}{u} &= \int xe^x dx = (x-1)e^x + C.\end{aligned}$$

Bernoulli equations

Example (Solve $\frac{dy}{dx} - y = xy^2$)

Rearranging the result from the preceding slide,

$$\frac{1}{u} = (1 - x)e^x - C, \quad y = ue^x = \frac{1}{1 - x - Ce^{-x}}.$$

This formula applies on intervals where its denominator is nonzero.

We divided by u^2 . The solution $u \equiv 0$ was therefore excluded from the calculation. It gives the additional solution

$$y(x) \equiv 0.$$